



Training vs Inference: Two Faces of AI Workloads

Understanding the distinct requirements of training and inference workloads is crucial for building efficient, scalable AI systems. This presentation explores how to architect infrastructure that addresses both phases effectively.

Two Core Phases of AI Workloads

AI systems operate in two distinct phases, each with unique infrastructure requirements:

1

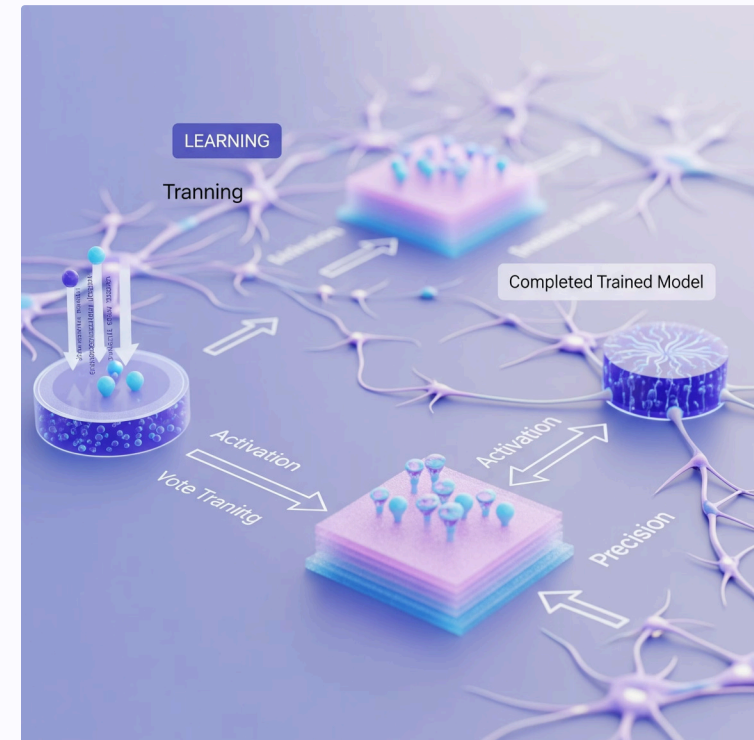
Training

The learning phase where models are taught from data, requiring intensive computation resources over extended periods

2

Inference

The deployment phase where trained models make predictions on new data, demanding low latency and high availability



Successful AI systems require strategic investment in both phases, with different optimization goals for each.



Training Workloads: Building AI Intelligence

Compute-Intensive

Requires massive parallel processing power from GPU/TPU clusters for weeks or months of runtime

Data-Hungry

Processes enormous datasets with high-throughput storage systems and data pipelines

Accuracy-Focused

Optimized for model quality and generalization ability rather than operational efficiency

Example: Training GPT-4 likely required thousands of GPUs processing trillions of tokens over many months at a cost of tens of millions of dollars

Inference Workloads: Delivering AI Value



Low Latency

Users expect real-time responses measured in milliseconds, not seconds



High Scalability

Must handle traffic spikes from thousands to millions of concurrent requests



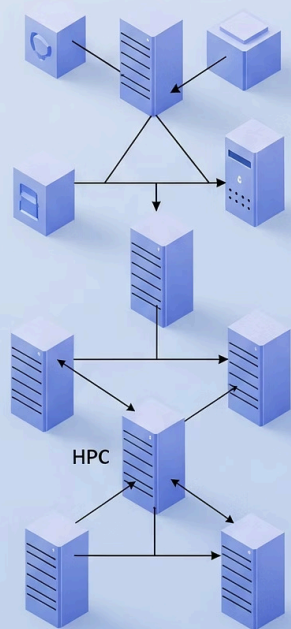
Deployment Flexibility

Runs on diverse hardware from cloud GPUs to edge devices to mobile phones

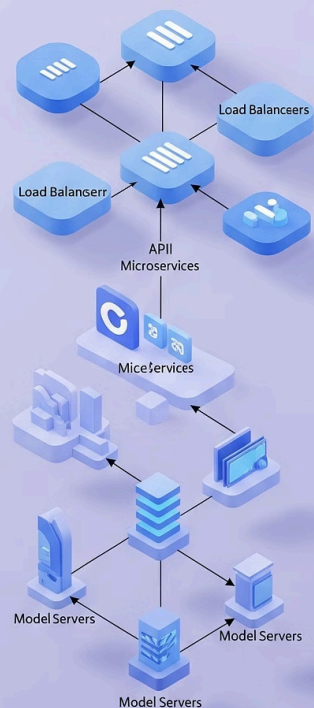


Example: ChatGPT processes over 10 million queries daily, each requiring real-time response while maintaining consistent performance.

Training Training



Inference for Inference Infrastructure



Infrastructure Differences

1

Training Infrastructure

- Distributed training clusters with high-speed interconnects
- Petabyte-scale storage with extreme throughput
- Expensive but used less frequently
- Tools: PyTorch DDP, Horovod, DeepSpeed

2

Inference Infrastructure

- Optimized serving systems with autoscaling capabilities
- API gateways and load balancers for traffic management
- Continuous operation with direct user impact
- Tools: Triton Inference Server, TorchServe, TensorRT



Infrastructure decisions made early in development can significantly impact both cost structure and system capabilities in production.

Why Both Matter: The Complete AI System

The most successful AI systems excel at both training **and** inference:

- Training creates the **intelligence** and capabilities
- Inference delivers the **value** to end users
- Optimizing both reduces costs and environmental impact
- Real-world success depends on efficient inference at scale

As AI infrastructure architects, our challenge is building systems that support this complete lifecycle while maintaining flexibility for future advancements.



"Great AI products aren't just about model quality—they're about delivering that quality reliably and efficiently to users at scale."